

EXAM 7 – SPRING 2013

10. (4 points)

Given the following information (\$000) as of December 31, 2012:

<u>Accident Year</u>	<u>Earned Risk Pure Premium</u>	<u>Adjusted Premium</u>	<u>Aggregate Reported Loss</u>	<u>Aggregate Loss Report Lag</u>
2010	9,200	9,200	3,600	70%
2011	8,500	7,400	2,500	50%
2012	10,000	10,000	4,200	30%

a. (1.5 points)

Use the Stanard-Bühlmann method to estimate the IBNR for accident years 2010 through 2012 combined.

b. (2 points)

Use a credibility-weighted combination of the Stanard-Bühlmann and chain ladder estimates to calculate a total IBNR estimate for accident years 2010 through 2012 combined. Assume credibility is a linear function of the report lag with a credibility factor of 0.35.

c. (0.5 point)

Briefly describe one advantage and one disadvantage of the Stanard-Bühlmann method as compared to the chain-ladder method.

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Question 10 Sample Answer

Solution 1

a)

AY	Used premium = (Adjusted prem) × (Report Lag)
2010	$9200(0.7) = 6440$
2011	3700
2012	<u>3000</u>
	13140

$$\text{ELR} = \frac{\Sigma \text{ Reported losses}}{\Sigma \text{ Used Premium}} = \frac{3600 + 2500 + 4200}{13140} = 0.784$$

AY	IBNR = (% unreported) × (Adjusted prem) × ELR
2010	$(1 - 0.7)(9200)(0.784) = 2164$
2011	$(1 - 0.5)(7400)(0.784) = 2901$
2012	<u>5488</u>
	<u>10,553</u>

b)

AY	LDF	IBNR = Reported loss × (LDF - 1)	Credibility	Cred. wtd IBNR
2010	$1/0.7 = 1.429$	$3600(1.429 - 1) = 1543$	$0.35 \times 0.7 = 0.245$	2,012
2011	$1/0.5$	2500	$0.35 \times 0.5 = 0.175$	2,831
2012	$1/0.3$	<u>9800</u>	0.105	<u>5,941</u>
				<u>10,784</u>

Cred. Weighted IBNR = $Z \times [\text{Chain ladder IBNR}] + (1 - Z)[\text{Stannard-Buh...IBNR}]$
Where $Z = 0.35 \times \text{Reporting lag}$

c) Advantage: more stable at less mature years

Disadvantage: requires many adjustments to earned premium that could be time consuming or difficult

Solution 2

a)

	<u>Used Up Premium</u>	<u>Unpaid = ELR × Adj Prem × (1 - Loss Lag)</u>
2010	6440	2164
2011	3700	2901
2012	3000	<u>5488</u>
		<u>10,533</u>

$$ELR = \frac{\Sigma Rep Loss}{\Sigma Used Up Premium} = 0.784$$

b)

	Unpaid CL = $\frac{Loss}{Loss Lag} \times (1 - Loss Lag)$	Cred Factor = $\rho_i \times 0.35$	Cred Weighted = $CL \times (z) + SB \times (1-z)$
2010	1543	.245	2012
2011	2500	.175	2831
2012	9800	.105	5941
			10,781

- c) The SB method uses a expected loss ratio, resulting in a more stable estimate of IBNR.
The CL method is more responsive to emerged loss experience than the S-B method.

Examiner Comment

This problem was fairly straightforward though calculation intense. Calculation errors were very common. For parts a) and b), several candidates forgot to provide the total IBNR as instructed in the question, but just provided it by year. For part c, a common error was providing an advantage of the Stanard-Bühlmann method relative to a method other than the chain ladder method.